

Eternals: Tackling The Large Scale Water Whirlpools

Gerardo Aguilera
gaguilera@wetafx.co.nz
Weta FX
Wellington, New Zealand

Kevin Blom
kblom@wetafx.co.nz
Weta FX
Wellington, New Zealand



Figure 1: Final rendered shot showcasing large scale water whirlpools

ABSTRACT

In Eternals third act we were faced with the challenge of creating extremely large water whirlpools that would spread for kilometres wide and deep. It was imperative that we approached this in an optimal way that would allow artistic control, reasonable turnaround time whilst maintaining a plausible and realistic look. This article focuses in the methodology used and how we took advantage of our in-house package Loki to achieve the results.

CCS CONCEPTS

• Computing methodologies;

KEYWORDS

FX, water, fluids, FFT Spectrums, simulation, open ocean

ACM Reference Format:

Gerardo Aguilera and Kevin Blom. 2022. Eternals: Tackling The Large Scale Water Whirlpools. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH '22 Talks)*, August 07-11, 2022. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3532836.3536267>

1 INTRODUCTION

Loki is Weta FX's distributed in-house simulation framework built for robust simulation of fluid, rigid, and deformable objects with emphasis on fidelity, coupling, and representation transitions across multiple elements. At its core Loki executes node graphs in various

host applications to load data, manipulate data, and send it somewhere. This can be either back into the host data or to the renderer. Loki is used also as our render procedural, it delivers volumetric and particle elements to renderers via what we refer to as a render graph. A render graph is a Loki graph executed at render time for delivering data, cached or live, to either our offline in-house renderer Manuka or our in-house real time viewport renderer Gazebo. Generally, it does not include a solver but will include loading data, manipulating it, and preparing it for the renderer.

2 PRODUCTION

In Eternals when we reach the third act, the characters face the challenge of the rising of the Celestial Tiamut. Tiamut has been dormant for centuries at the core of the Earth and with his awakening, his body begins to rise in the middle of the ocean. We get to see his head and hand triggering the formation of water whirlpools. In some shots this water whirlpool would extend for kilometers. The scale was something we knew would become the biggest challenge to sell since ultimately it was so “unreal” it was difficult to conceptualize.

The start of the process for most FX work usually begins with finding real life reference but for obvious reasons we couldn't find much that would help with our brief. The largest real whirlpool we could find was the Nordic Maelstrom which is about 100x smaller than what we would need to make for Tiamut's rising. This footage did however spark some early tests for trying to simulate the whirlpool which started to look quite promising. From here we moved on to exploring multiple options by manipulating the simulation, from creating a flip simulation at a 1/1000th of a scale, to trying to retine and slow down heavily the simulation. With all these different attempts, we kept facing mainly two problems, the turnaround time was not really “production-realistic”, and we struggled to find the right balance between both simulating something

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '22 Talks, August 07-11, 2022, Vancouver, BC, Canada

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9371-3/22/08.

<https://doi.org/10.1145/3532836.3536267>

“realistic” that would look believable and plausible but at the large scale it needed to be.

The core of our approach was to move away from a simulated tank, and we turned to our already established and proven open ocean workflow using Loki. This workflow is typically only used for creating realistic procedural oceans. We started exploring the idea if this workflow could be adapted in some way to form the basis of our whirlpool.

Our open ocean workflow usually involves stacking multiple FFTs where individual FFTs can be instantiated to points, each point giving control over important features such as the wave orientation, coverage and timing. We can also control some of the wave properties (such as amplitude) using volume masking, meaning that in a typical ocean scenario we can paint in calmer areas or fade out swells.

The idea we had was to exploit the FFT splatting and masking workflow by orienting patches and masking them in such a way that we could start to create a whirlpool like behaviour. Traditionally with Houdini’s open ocean for applying FFT spectrums to the water surface would give us a uniform direction of the waves which would not work with the circular nature of the whirlpools. Loki gave us the flexibility to “guide” the direction of our waves to follow the circular motion of the whirlpool. We accomplish this by scattering points on our surface, from there each point represents a section of the FFT to splat down. Each point contains a direction for the FFT spectrum waves and a falloff, so that when we had enough overlapping spectrums splatted it blended to create a seamless and interesting FFT pattern.

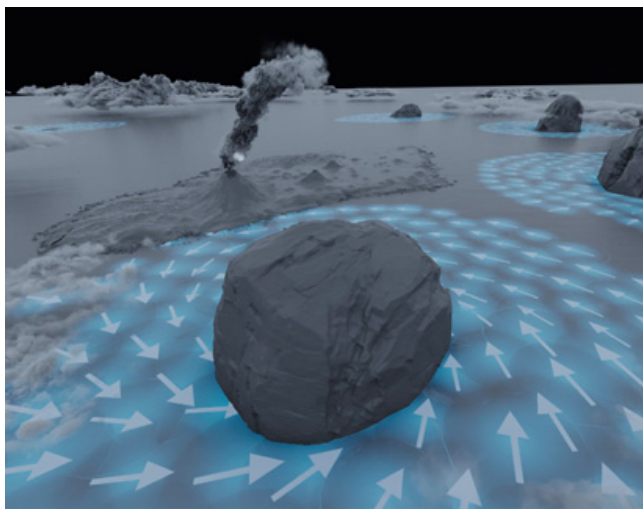


Figure 2: Visual demonstration of the guided FFT Spectrums splatted per point.

The first tests looked promising in that we had the appearance of water swirling but the missing component was the funnel shape that arises from a whirlpool.

An important feature that Loki provides us in the render graphs is the ability to composite volumes. It consists of a volume made of a graph of operations, similar to a shading network. The operations are evaluated on-demand per sample and allow for effective

modification of read-only data and mixing multiple volume sources together. We use these operations extensively in render graphs with other workflows as well.

The volume compositing allowed us to easily add in a heightfield to offset our implicit ocean plane. This approach gave us more control over the whirlpools shape and size and we didn’t have to generate a mesh in each iteration for the final render.

To accomplish this, we started with a flat geometric plane in houdini and using custom deformers we deformed the geometry into the basic whirlpool shape. We then applied an orbital motion and the suction towards the centre using VEX code. The main basis of the deformation involves the use of custom noise patterns which are twisted and layered in such a way that we were able to create a convincing whirlpool shape and motion. To achieve the shape of the funnel we used a distance falloff to the centre of the whirlpool that was remapped to give the appropriate profile.

We then converted the deformed whirlpool geometry into a heightfield which is used in the render graph to give us the basic funnel shape, then on top of that we layered on FFT spectrums to add the water waves.

At the end we had the two components, the implicit water surface with the FFTs and the deformed funnel shape of the whirlpool. In order to seamlessly combine these, we used volume compositing inside the Loki render graph. Although for some shots we still had to create hero simulations in some areas, with Loki’s render graphs we were able to blend the hero simulation surface with the implicit surface seamlessly.

The final additions were to add secondary water components – splashes, mist, foam and aeration. To achieve the final look, it required a layering approach using a combination of simulated, procedural and plate elements. For the simulated elements we were able to run guided simulations using our procedural whirlpool surface to drive the emission and dynamics. We also provided comp with a coordinate space representation of our water surface which allowed them to dress in some final foam elements using plate footage.

3 FUTURE WORK

The Simulation RnD team are in an ongoing task of improvement and development of Loki. Some of the focuses planned for the future are in incorporating more solvers into the framework such as an atmospheric cloud simulator. There is also focus on improving the performance of the current simulations, for example employing a wider use of spatial adaptivity for fluids and varying topology for meshes, reducing the amount of data to be processed in regions where high resolution is not necessary. There is also a known bottleneck in resolving heavy collisions between all kinds of geometries and representations, the Simulation team would like to explore ways to expedite this step of the solver.

ACKNOWLEDGMENTS

We like to acknowledge the FX team in Weta FX in particular to those who contributed in the making of Eternals and the Simulation RnD team in Weta Digital for their work, development, testing and contributions to Loki.